

Understanding Key Stakeholders' Perspectives Towards Artificial Intelligence in Home Care Work



Ian René Solano-Kamaiko, MS¹, Michael Dicipinigaitis, BS², Melissa Tan, MS¹, Ariel Avgar, PhD³, Aditya Vashistha, PhD³, Nicola Dell, PhD¹, and Madeline R. Sterling, MD, MPH, MS²

¹Cornell Tech, New York, NY, USA; ²Weill Cornell Medicine, New York, NY, USA; ³Cornell University, Ithaca, NY, USA

ABSTRACT

BACKGROUND: Older adults are increasingly relying on home health aides and attendants (HHAs) to support their care needs and enable them to age in place. In response to workforce shortages and rising care demands, AI systems are often touted as potential solutions. However, little is known about how stakeholders responsible for implementing these systems perceive AI's potential benefits and risks.

OBJECTIVE: To examine the perspectives of HHAs and other key stakeholders on deployment of AI technologies in home care work.

DESIGN AND PARTICIPANTS: A qualitative study was conducted with English-speaking HHAs, home care agency leaders and staff, worker advocates, clinicians, and technology company leaders and staff. Purposive sampling and snowball sampling were used to identify and recruit participants. Semi-structured interviews, including AI scenarios, were conducted.

MAIN MEASURES: Interviews were analyzed using a rigorous three-stage qualitative analysis process: structural coding, inductive sub-code generation, and thematic analysis.

KEY RESULTS: A total of 43 participants across five stakeholder groups participated. The mean (SD) age was 44.6 (11.2) years, 23 (63.9%) identified as women, 35 (97.2%) had completed at least some college education, and 16 (44.4%) reported having no or low knowledge of AI. Four major themes emerged: (1) potential benefits of AI to enhance patient care, strengthen HHA engagement, and improve organizational efficiency; (2) risks of AI eroding patient care, harming provider-patient relationships, and worsening working conditions for HHAs; (3) challenges related to data quality, privacy protections, and AI literacy among HHAs; and (4) the need for timely, equitable AI partnerships and governance.

CONCLUSIONS: Participants recognized both the promise and risks of AI in home care. Findings highlight the need for early, inclusive governance frameworks and labor protections to ensure that AI adoption supports, rather than undermines, care quality, workforce sustainability, and patient-provider relationships in the home care sector.

KEY WORDS: home health aides; home care services; artificial intelligence; health services research

J Gen Intern Med

DOI: 10.1007/s11606-026-10782-z

© The Author(s) 2026

INTRODUCTION

The United States (U.S.) is facing a growing home care crisis. As the population of older adults increases, many are experiencing complex medical needs and cognitive decline, often relying on a limited and overstretched workforce of paid home health aides and attendants (HHAs) to support their care and enable aging in place¹. Despite their essential role, HHAs—who are predominantly women from racial and ethnic minority backgrounds²—earn low wages³, receive insufficient training and recognition⁴, are undervalued by the healthcare system^{5,6}, and often work in isolation^{7,8}. In response to persistent workforce shortages and rising care demands, artificial intelligence (AI) systems are frequently promoted as potential solutions, with the promise of improving operational efficiency, reducing costs, and alleviating strain on the home healthcare system⁹.

As AI technologies begin to be deployed across the home care sector¹⁰, supporting functions such as staffing, scheduling and coordination, regulatory compliance, predicting patient outcomes, and monitoring worker performance¹¹, their integration is poised to have both intended and unintended consequences. While such tools may improve care coordination and workforce engagement, they risk introducing new burdens for workers and may erode the relational and hands-on aspects that are central to home care work.

Across the healthcare sector, AI adoption has increased from 5.9% in 2023 to 8.3% in 2025, with a 481.5% acceleration in the rate of adoption after late 2024, indicating a sharp inflection towards rapid uptake¹². Reflecting this broader trend, AI use in home care is also expanding, and which is reflected in the growing use of AI-enabled home-based technologies, including wearable monitoring devices and ambient in-home sensor systems that support health tracking and early detection of clinical changes¹³. Furthermore, platforms enabled by AI can perform tasks such as automatically adjusting caregiver schedules in real-time based

Prior Presentations: An earlier version of this work was presented at the ACM CHI conference on Human Factors in Computing Systems, which was held from April 26th to May 1 st, 2025.

Received May 15, 2026

Accepted August 17, 2026

Published online: 22 September 2026

on patient acuity and geographic proximity, while in-home sensors and wearables can continuously track movement patterns, sleep, and vital signs that integrate with AI to provide alerts that caregivers can address. Other tools may generate individualized care plans from aggregated patient data and even serve as virtual assistants to guide patients through daily care tasks, symptom reporting, and communication with care teams^{9,10}.

Yet despite new innovations and this rapid uptake of AI use, as well as its implications for the workforce and their patients, there remains limited understanding of how HHAs and other key stakeholders in the long-term care industry perceive these systems. Specifically, little is known about their perspectives on the benefits and challenges of AI, expectations regarding stakeholder roles and responsibilities, and concerns related to data collection, accountability, and governance. To address these gaps, we conducted a qualitative study with HHAs and other key stakeholders including home care agency leaders and staff, worker advocates, clinicians (physicians and nurses), and technology company leaders and staff to understand their experiences and perspectives on current and near-future AI deployment in home care work.

METHODS

Setting and Study Population

We conducted a qualitative study between July 22nd, 2024, and April 18th, 2025. Using purposive sampling, we recruited participants across five key stakeholder groups identified through prior research: HHAs, home care agency leaders and staff, worker advocates (HHA labor advocacy organizations and unions), physicians and nurses (with experience in home care work, clinical care of older adults, and/or home care research), and technology company leaders and staff^{8,14}. Here we use the term “stakeholder” to refer to key groups with direct experience in or influence on home care delivery, including those providing care, organizing or advocating for the workforce, shaping clinical practice, or developing and implementing relevant technologies. Participants were recruited through purposive sampling from prior and ongoing research collaborations and the research team’s professional networks—meaning that in some cases, interviewers had pre-existing relationships with interviewees prior to conducting interviews. Snowball sampling was used to expand recruitment efforts. We invited 65 individuals to participate; 43 agreed, including 12 recruited through snowball sampling. Eligible participants were English-speaking adults aged 18 years or older who were actively employed in one of the designated stakeholder groups. All participants were based in the U.S., with most based in major urban areas in the northeast, particularly within the New York tri-state region. Participants represented a range of organizations

within each stakeholder group, including multiple home care agencies, hospital networks, labor organizations, and technology companies. Recruitment was initiated via email using a standardized script that outlined the study’s objectives and voluntary nature. Interested individuals were screened for eligibility and scheduled for an interview. All procedures were approved by the Cornell University Institutional Review Board. Informed consent was obtained in accordance with the Common Rule and our Institutional Review Board. Participants received a \$25 electronic gift card as compensation for their time. This study follows standard reporting for qualitative research methods (Supplemental File—COREQ).

Data Collection

Three researchers (I.R.S.K., M.T., N.D.), trained in qualitative methods conducted semi-structured interviews via Zoom video conferencing software using topic guides. Most interviews were 1-on-1, with one exception: a group interview was held with staff members in a variety of roles from a single major union, as they were the only organization to request this group format. Interviews lasted approximately 60 min. The topic guides were informed by prior research conducted by the research team^{8,14}, internal AI domain expertise, secondary research on commercially available AI tools in home care^{15–17}, and literature on technology adoption in frontline care work^{18–21}.

Participants were asked about their occupational backgrounds, work experiences, and AI understanding. Participants were then presented with fictional vignettes that explored using an AI tool to (1) facilitate HHA-patient matching, (2) monitor HHA-patient interactions in the home, (3) improve HHA care coordination tasks (Table 2). Vignettes were selected as a methodological tool to ensure that conversations across stakeholder groups—who varied in their familiarity with AI—were grounded in a consistent and realistic depiction of how AI might be implemented in home care. The vignettes were based on real AI products marketed for home care^{15–17} and were carefully worded to encourage balanced reflection on the potential benefits, risks, challenges, and tensions associated with AI use. Each participant was assigned two of the three vignettes (independent of stakeholder role) to ensure interviews remained within the allotted time. Participants read the vignettes independently during the interview, with the interviewer available to clarify any questions. Interviews concluded with broader questions about governance and control of AI. All guides shared a core set of questions with minor wording adjustments tailored to each stakeholder group. For participants from technology companies, the interview guide underwent more substantive modifications. Given their deep familiarity with AI, the vignettes were omitted and questions instead focused on their company’s AI products and initiatives.

We collected self-reported demographic data, including age, gender, education level, and employment history. We

Table 1 Participant Characteristics

Characteristics	No. (%)
Age, mean (SD)	44.6 (11.2)
Years of professional experience, mean (SD)	11.3 (7.8)
Female, mean (SD)	23 (63.9)
Knowledge of AI by participant type, <i>n</i> (%)	
Home health aides	
No knowledge	1 (9.1)
Low knowledge	10 (90.9)
Home care agency upper management	
Moderate knowledge	6 (100)
Home care agency lower management	
Low knowledge	3 (75)
Moderate knowledge	1 (25)
Work advocates	
Low knowledge	1 (7.1)
Moderate knowledge	4 (28.6)
High knowledge	2 (14.3)
Technology company leaders and staff	
High knowledge	3 (100)
Nurses	
Low knowledge	1 (50)
High knowledge	1 (50)
Physicians	
Moderate knowledge	2 (66.7)
High knowledge	1 (33.3)

Missing data: seven participants (all worker advocates) were interviewed in a group setting where we were unable to capture this information. Data were self-reported using open-ended questions. Responses for sex, educational level, and knowledge of AI were subsequently categorized (as shown). We use the term “stakeholder” to refer to key groups with direct experience in or influence on home care delivery, including those providing care, organizing or advocating for the workforce, shaping clinical practice, or developing and implementing relevant technologies

also asked about participants’ familiarity with AI during the interview and categorized their verbal responses into four groups: *No Knowledge* (no understanding of basic AI principles; cannot identify AI-driven technologies), *Low Knowledge* (limited understanding of how AI works; can recognize examples such as ChatGPT), *Moderate Knowledge* (understand key AI concepts; understand that AI uses data for training to improve its accuracy), and *High Knowledge* (AI training; typically courses in computer sciences or relevant fields; or experience building AI systems).

Data Analysis

Interviews were audio recorded and transcribed using NoScribe²², an open-source AI transcription tool run locally on lab computers by two members of the team (I.R.S.K., M.T.). Transcripts were coded by two investigators (I.R.S.K., M.T.) using Google spreadsheets and ATLAS.ti²³ software for qualitative data analysis. We followed Kuckartz’s²⁴ three-stage process for qualitative data analysis: (1) structural coding of high-level topics; (2) inductive generation of sub-codes within these topics; and (3) thematic analysis within and across topics. Structural coding, a deductive first-cycle

coding method²⁴, was guided by the research questions and themes in the interview topic guides. This involved multiple passes through the transcripts to assign codes aligned with these predetermined topics. To develop a comprehensive understanding of the data, within each topic we conducted inductive, open coding to generate sub-codes, yielding a total of 223 sub-codes. We then used Braun and Clarke’s approach to thematic analysis²⁵ to identify and refine key themes (see Box 1). The data within each theme were linked to individual participants, enabling us to identify differences by participant role when present. Theme development was supported by three team members (M.D., N.D., M.R.S.) who did not code interviews but brought relevant content expertise. Throughout the analytic process, the research team employed a collaborative qualitative approach, meeting regularly to review coding, resolve discrepancies, and ensure consistency through agreement.

RESULTS

A total of 43 participants were interviewed, including 11 HHAs, 10 home care agency leaders and staff, 14 worker advocates, 5 clinicians (physicians and nurses), and 3 technology company leaders and staff (Table 1). Among the 36 participants for whom demographic data were available, the mean (SD) age was 44.6 (11.2) years, 23 (63.9%) were women, and 35 (97.2%) had completed some college or higher education. Thematic analysis resulted in four major themes with accompanying subthemes (Box 1), which are presented alongside representative quotations

Box 1. Themes and Subthemes

Major Themes and Subthemes Summarizing Key Stakeholders Perspectives Towards Current and Near Future Use of AI in Home Care.

Theme 1: The benefits of AI: potential to enhance patient care, strengthen HHA workforce engagement, and improve organizational efficiency

Subthemes

- Potential for AI to improve documentation and support caregivers in the home
- Potential to improve HHA engagement and retention
- Efficiency gains for advocacy organizations and agencies

Theme 2: Potential risks of AI to Erode patient care, provider-patient relationships, and working conditions for HHAs

Subthemes

- Potential for AI to harm the quality of patient care delivered in the home
- Concerns about the increasing data collection burden on HHAs and agency staff
- Risk of AI exacerbating workforce shortages in an already-strained labor market

Theme 3: Challenges around data responsibilities, privacy protections, and AI literacy

Subthemes

- Responsibility for data quality and freshness
- Concerns about data privacy and surveillance for HHAs and patients
- Limited HHA understanding of AI can hinder informed advocacy efforts

Theme 4: A Need for timely equitable AI partnerships and governance in home care

Subthemes

- Early-stage AI offers opportunities for organized labor
- Need for organizational capacity building around AI
- Stakeholders want AI governance and regulation in home care

Theme 1: The Benefits of AI: Potential to Enhance Patient Care, Strengthen HHA Workforce Engagement, and Improve Organizational Efficiency

Potential for AI to Improve Documentation and Support Caregivers in the Home. Participants perceived that AI systems could help HHAs and family caregivers provide home-based care. Participants thought that AI systems offered new ways to document symptoms and patient data in the home, or extract important care information. One HHA said:

[The AI] is able to record things that were done and things that weren't done in order to let the care team know how to improve the patient's care.

Clinicians concurred with AI's ability to capture clinically relevant data in the home. They also perceived that AI might improve coordination across different caregivers. A participant said:

Maybe one aide reported issues with incontinence and another aide when there was a fall. Different things happened on different watches and maybe not all of it got back to a family member that's coordinating things. ... [An AI system] could break down those barriers of information so different individuals are on the same page.

Potential to Improve HHA Engagement and Retention. Some participants—particularly those working for technology companies offering these tools—further described how AI might support home care agencies with engaging and retaining their workforce by, for example, increasing access to work opportunities and improving the quality of job matches. A technology company participant said:

[Agencies] want to keep [HHAs] happy ... [Technology companies] need to do a better job of figuring out what is going to be helpful for [HHAs] ... they're only going to stay if we can get them working and keep them happy.

Efficiency Gains for Advocacy Organizations and Agencies. The majority of home care agency participants

also emphasized that AI systems offer important efficiency gains for their organizations. A staff member at a home care agency currently employing an AI-driven shift-matching system said:

We've seen the benefit ... it is easier for us to shift match much more efficiently.

Furthermore, multiple worker advocates noted that their organizations could leverage AI systems internally to support overextended organizers and improve response times for workers who have inquiries about labor conditions, compensation, and time off. A worker advocate described:

Someone might have a relatively simple question about their contract. Like 'how many paid vacation days do I get after six years'... People are using [AI] to respond quicker instead of [the worker] having to interact with a busy organizer.

Theme 2: Potential Risks of AI to Erode Patient Care, Provider-Patient Relationships, and Working Conditions for HHAs

Potential for AI to Harm the Quality of Patient Care Delivered in the Home. Despite its potential benefits, many participants, particularly clinicians, expressed concerns about possible degradation of patient care, including how AI might harm patient-provider relationships. A physician said:

I worry that [AI] can dehumanize care ... there's something about the connection between the provider and the care recipient who could feel alienated by the intervention of machinery ... I can imagine a world in which we're supposed to trust AI so we distrust each other and that might be corrosive.

Additionally, participants worried that AI use would increase administrative workload, particularly related to data collection, potentially reducing the time HHAs spend on direct, human-to-human care:

The person that's left with [more] administrative [work] would most likely be the aide... to log things they previously didn't have to ... Now, they're spending all their time on the iPad filling out forms and not with the client.

Concerns About the Increasing Data Collection Burden on HHAs and Agency Staff. All stakeholder groups, particularly home care agency staff and worker advocates, perceived that AI use might increase burdens on HHAs and agency staff. They pointed out that AI systems may require significant amounts of HHA, patient, and clinician data to be manually inputted to function properly and expressed concern that HHAs and agency staff would bear primary responsibility for

accurately and responsibly inputting, maintaining, updating, and removing this information. Based on prior experience implementing a non-AI technology application at their organization, an agency staff participant anticipated that AI systems would pose similar data collection challenges:

It was a huge undertaking. Hundreds of hours to manually input every aide, every patient, every calendar ... a team needs to keep all the information accurate.

Risk of AI Exacerbating Workforce Shortages in an Already-Strained Labor Market. Participants felt that the increased administrative burdens for HHAs might exacerbate workforce shortages by diminishing job quality and ultimately leading to resignations, an outcome that runs counter to the benefits often claimed by technology companies. A worker advocate said:

That administrative burden takes out ... what might be enjoyable about the work ... You remove the relationship aspect or things that are still holding people to these [jobs].

Participants further noted that AI technologies would likely center on the needs of home care agencies, which might lead to wage suppression for HHAs, further deteriorating an already challenging labor environment. A clinician said:

If you're ... investing in AI tools ... the place where this will have the most traction is where there's incentives to maximize profits and minimize wages.

Theme 3: Challenges Around Data Responsibilities, Privacy Protections, and AI Literacy

Responsibility for Data Quality and Freshness. As the prior theme suggests, the performance of AI-driven systems is often predicated on the quality and freshness of the data they ingest. When asked who would be responsible for maintaining these heterogeneous datasets, and performing quality control, which can require detailed information such as patients' care requirements and HHAs' skills and training, a technology company participant said:

It comes back [to]: the system's only as good as the information you put in it ... at the end of the day, who is held accountable for that?

Concerns About Data Privacy and Surveillance for HHAs and Patients. The majority of clinicians raised questions

about patient privacy in the home and how personal information would be used by AI systems. A clinician noted:

It is a means of surveillance ... you'd want the care recipient to be able to opt out or to say, 'I don't want [the AI] telling someone if this happens ... I don't want it shared'.

Furthermore, a worker advocate participant emphasized the importance of HHAs understanding how their data are being used:

Making sure that people know what's happening with the data. Can they opt out of the data ... understanding what the data is being used for. They should at least have a sense of it.

Although clinicians and worker advocates emphasized the importance of opt-out functionalities as a safeguard, home care agency leaders and staff noted these options as impractical or unrealistic. Moreover, HHA participants employed by these agencies expressed a perceived lack of agency in these decisions. A HHA noted:

If my agency tells me, we're going to put AI in your patient's home, I cannot say no.

Limited HHA Understanding of AI Can Hinder Informed Advocacy Efforts. HHAs expressed interest in participating in labor contract negotiations and having meaningful input in AI governance structures, but participants worried that HHAs' limited understanding of AI could curtail such advocacy efforts, noting that HHAs may struggle to articulate concerns or preferences in ways that could inform labor organizing or policy efforts. A worker advocate said:

Do [HHAs] want to prioritize a bigger raise or better job protections? ... It's our responsibility to educate our members about what AI is. We need to have a better plan to educate [HHAs] so we can say to them that you have to ... tell us what you want in your contract.

Theme 4: A Need for Timely Equitable AI Partnerships and Governance in Home Care

Early-Stage AI Offers Opportunities for Organized Labor. Our data suggests the home care industry stands at a critical inflection point with respect to AI. Given that AI adoption in home care is nascent, participants emphasized the importance of proactively establishing operational guidelines and labor protections before these technologies become fully entrenched. A worker advocate said:

We are low on the disruption list. So we can take a bit of space and try to set an agenda, which isn't naive ... manipulate the technologies for [our] labor goals.

Need for Organizational Capacity Building Around AI. Despite this optimism, some participants expressed concern about the capacity of their organizations to engage with AI technologies due to limited internal expertise. An advocate said:

Could we be actually ourselves involved? ... we don't have a lot of computer science graduates ... it might be that we've got to figure out how to retool ourselves.

Stakeholders Want AI Governance and Regulation in Home Care. Nearly all participants wanted clear governance and regulatory frameworks for AI in home care. Many emphasized the importance of ensuring that all stakeholders have input in the design and oversight of these systems. A clinician said:

You want to have a diversity of perspectives. Certainly you want care recipients, their family members, and ... different people from the care team.

DISCUSSION

To our knowledge, this is one of the first studies to examine key stakeholder perspectives on current and near-future AI deployment in home care. Participants identified potential benefits of AI, including enhanced patient care through improved documentation, greater organizational efficiency, and expanded job opportunities for HHAs. However, they also noted potential challenges, such as reduced human interaction, erosion of trust between providers and patients, and increased data burdens on HHAs. Given that AI adoption in home care is still nascent, these findings highlight the need to establish equitable partnerships and governance structures before these technologies become fully entrenched.

This study adds to literature examining the potential impacts of AI in the long-term care workforce^{18–20,26,27}. Previous research has primarily examined AI within institutional care settings, such as nursing homes and residential care facilities. For example, Barbosa Neves et al.²⁷ found that nursing home staff were concerned about AI dehumanizing care and noted a disconnect between technology developers and care practitioners regarding trust in these systems. Beyond institutional care, Wang et al.²⁸, examined chronic patients' perspectives on AI-based home care, highlighting the importance of personalized interfaces for older adults, social connectivity, integration with existing technologies, and needs for ethical guidelines and affordability.

Our study adds to this literature by examining roles for AI in home care contexts. While we similarly found concerns about AI's potential to harm patient-provider relationships

and foster distrust, our findings suggest that AI may also exacerbate existing vulnerabilities within the HHA workforce. For AI systems to function effectively, high-quality data must be collected, maintained, and validated, a responsibility likely to fall on HHAs—the same workers who provide hands-on care. This added responsibility may be taxing physically, emotionally, and financially, particularly given that HHAs often work in under-resourced home environments without the training, support, or tools available in institutional settings, all while receiving some of the lowest wages in healthcare. Our findings further suggest that these data-related demands could worsen workforce shortages in an already strained labor market. Participants noted that these tasks can become burdensome, reducing human interaction and diminishing the relational aspects of care—elements essential to providing high-quality care and contributing to the fulfillment HHAs find in their work.

Our findings also suggest that deploying AI in the home may be more complex than other care settings. Unlike institutional care facilities, the goal of long-term care at home (i.e., Medicare skilled home health or Medicaid-funded personal care) is to help older adults maintain or improve their symptoms and functioning while remaining at home. HHAs are a highly distributed workforce, often working in isolation in patients' homes with limited oversight, support, and connection to peers^{7,8}. This work occurs in an intensely intimate environment where HHAs typically have less power than patients, families, clinicians, and other stakeholders^{14,29,30}. Receiving care at home also presents unique challenges regarding patients' and family caregivers' normative expectations of data privacy. While patients often accept continuous monitoring in hospitals or nursing homes to diagnose or monitor their conditions³¹, our findings suggest that similar monitoring in the home may not be as readily accepted. Homes are personal, intimate, and non-medicalized spaces that may require a more sensitive approach to data collection, including clear consent from patients and families and the ability to opt out of monitoring practices.

Lastly, our findings highlight a time-sensitive opportunity to shape the future of AI in home care through equitable governance. AI adoption in home care is in its early stages and stakeholders have a unique window to develop inclusive governance structures that center on those most affected—HHAs and patients. We found that HHAs often lack understanding of how their data are used within AI systems and expressed limited agency to decline participation, which may also impede their ability to engage in advocacy and contract negotiations. Advocacy organizations, including labor unions, can play a pivotal role by embedding AI-related labor protections into contracts, facilitating AI education, and fostering strategic partnerships with technology developers. Although home care is highly regulated under laws such as HIPAA³² and Electronic Visit Verification

Table 2 Interview Vignettes

Vignette 1

Imagine Suzie is a home care agency administrator. Her job is to match and manage multiple home care workers and patients. Her agency recently decided to use an AI tool called PairMe to help her match home care workers with patients. PairMe looks at a lot of information about the agency's business goals, home care workers, and patients. Using all this information PairMe makes a prediction. PairMe suggests Suzie pair Mary, a home care worker employed by the agency, with a patient in Upper Manhattan who was previously working with a different care worker

Vignette 2

John has recently been diagnosed with Alzheimer's Disease. He needs support throughout the day, yet often resists care, expressing anger and frustration with his caregiver Maria. John also often does not recognize Maria. The home care agency that Maria works for recently started using an AI tool called HearMe to better understand what is happening in patients' homes. HearMe uses audio speakers and microphones like Amazon Alexa and mobile phones to record audio in a patient's home. HearMe analyzes the audio to detect information like signs of distress, conflict, happiness, or other important information. In this case, HearMe alerted the agency staff that John is having intense interactions with Maria around the time she arrives every morning. HearMe also detects that John typically resists care during the morning but later in the afternoon, he tends to have more positive interactions with Maria. Based on this information, HearMe suggests shifting more challenging care activities to the afternoon when John tends to be less resistant. It also suggests that the agency give Maria extra training

Vignette 3

John has been diagnosed with Alzheimer's disease and needs daily support. His family has arranged for home care, and the agency managing his case now uses an AI tool called ChatMe. ChatMe helps improve care coordination by using information like caregiver conversations, care preferences, and medical records. It acts as a personal assistant, sending proactive check-ins and reminders, summarizing completed tasks, flagging potential concerns, and more, ensuring personalized and continuous care for John

requirements³³ for Medicaid-funded services, legislation specific to AI has yet to emerge, despite the World Health Organization's³⁴ recommendations. Given stakeholders' desire for governance and the industry's history of national safety standards, incorporating AI guidelines into existing home care policies may be a timely and practical pathway to ensure responsible and equitable AI adoption in this high-stakes setting.

Strengths and Limitations

Strengths of this study include its timeliness and a strong community-engaged approach. We recruited a diverse sample of participants and employed a rigorous, three-stage qualitative analysis process to systematically interpret the data. This study also has limitations. As a small-scale qualitative study in urban settings in the U.S., the generalizability of our findings to larger samples and other contexts, including rural or non-U.S. settings, remains unknown. We also recognize that although the vignettes used with participants were designed to minimize bias, they may have influenced participants' perspectives beyond their own experiences, which could have impacted our findings (Table 2). Furthermore, limitations inherent to qualitative interviews, such as potential recall bias, may be present. Also, it is important to consider the perspectives of older adults and their caregivers to more deeply understand how their preferences and needs intersect with AI in home care. Future research could involve these groups as well as employ ethnographic methods to examine real AI deployments in home care. Lastly, our study did not formally assess participants' current use of AI technologies in their work. Although, given the observed disparities in AI knowledge based on stakeholder groups, this potentially represents an important direction for future research.

CONCLUSIONS

In this qualitative study, key stakeholders identified opportunities and challenges associated with AI's use in home care. While AI tools may improve patient care, documentation, and workforce engagement, they risk eroding patient-provider relationships, increasing burdens on an already strained workforce, and raising concerns about data quality and privacy. As AI adoption in home care is in its early stages, there is a critical window for stakeholders to collaboratively develop equitable governance frameworks that prioritize the needs of HHAs and patients. Doing so will be essential to ensuring that AI supports, rather than undermines, care quality and workforce sustainability in this high-stakes sector.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11606-026-10782-z>.

Acknowledgements We would like to acknowledge the contributions of Joy Ming for their assistance with participant interviews and recruitment. We would also like to thank the study participants for sharing their time and experience to make this study possible.

Corresponding Author: Madeline R. Sterling, MD, MPH, MS; , Weill Cornell Medicine, New York, NY, USA (e-mail: mrs9012@med.cornell.edu).

Author Contribution ISRK, AV, ND, and MRS conceived the study. ISRK, MT, and ND collected study data. ISRK, MD, MT, AA, AV, ND, and MRS contributed equally and substantially to analysis, manuscript writing, review, and revision and approved the final version of the manuscript.

Funding This study was supported by the Innovation Resource Center for Human Resources (IRC4HR), alongside a Google Cyber NYC award, and funding from the CAROW Initiative on Home Care Work, and an American Heart Association grant (AHA#23SCISA1142170). The content is solely the responsibility of the authors and does not necessarily represent the official views of these organizations and entities. Additionally, the funders had no role in the design and conduct of the study; collection, management,

analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Availability The datasets generated and analyzed during the current study are not publicly available due to participant privacy, but aspects of the data including the topic guide and code book are available from the corresponding author on reasonable request.

Declarations

Ethics Approval and Consent to Participate All participants gave informed consent to participate in the study and were compensated for their time. All procedures were approved by the Cornell University Institutional Review Board.

Conflict of interest The authors declare that they do not have a conflict of interest.

Clinical Trial Number Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

REFERENCES

1. Markkanen P, Quinn M, Galligan C, Chalupka S, Davis L, Laramie A. There's no place like home: a qualitative study of the working conditions of home health care providers. *J Occup Environ Med*. 2007;49(3):327-337. <https://doi.org/10.1097/JOM.0b013e3180326552>.
2. PHI National. *Direct Care Workers in the United States: Key Facts 2024*. 2024. New York, NY: Paraprofessional Healthcare Institute; 2024. <https://www.phinational.org/resource/direct-care-workers-in-the-united-states-key-facts-2024/>. Accessed Jul 2, 2025.
3. Howes C. Home care: the fastest growing low-wage industry. *New Labor Forum*. 2015;24(2):98-105. <https://doi.org/10.1177/1095796015579692>.
4. Sterling MR, Dell N, Tseng E, et al. Home care workers caring for adults with heart failure need better access to training and technology: A role for implementation science. *J Clin Transl Sci*. 2020;4(3):224-228. Published 2020 Apr 6. <https://doi.org/10.1017/cts.2020.36>.
5. Hewko SJ, Cooper SL, Huynh H, et al. Invisible no more: a scoping review of the health care aide workforce literature. *BMC Nurs*. 2015;14:38. Published 2015 Jul 22. <https://doi.org/10.1186/s12912-015-0090-x>.
6. Sterling MR, Silva AF, Leung PBK, et al. "It's Like They Forget That the Word 'Health' Is in 'Home Health Aide'": Understanding the perspectives of home care workers who care for adults with heart failure [published correction appears in *J Am Heart Assoc*. 2018 Dec 14;7(24):e002256. doi:10.1161/JAHA.118.002256.]. *J Am Heart Assoc*. 2018;7(23):e010134. <https://doi.org/10.1161/JAHA.118.010134>.
7. Osakwe ZT, Osborne JC, Samuel T, et al. All alone: a qualitative study of home health aides' experiences during the COVID-19 pandemic in New York. *Am J Infect Control*. 2021;49(11):1362-1368. <https://doi.org/10.1016/j.ajic.2021.08.004>.
8. Sterling MR, Li J, Cho J, Ringel JB, Silver SR. Prevalence and predictors of home health care workers' general, physical, and mental health: findings from the 2014-2018 behavioral risk factor surveillance system. *Am J Public Health*. 2021;111(12):2239-2250. <https://doi.org/10.2105/AJPH.2021.306512>.
9. Home-Based Care Providers Turn to AI to Address Staffing Shortages and Improve Efficiency. *Home Care Association of America*. December 18, 2024. <https://www.hcaa.org/newsletters/home-based-care-providers-turn-to-ai-to-address-staffing-shortages-and-improve-efficiency>. Accessed May 4, 2026.
10. Klein J. The Coming AI Revolution in Home Care. *Forbes Business Council*. June 2, 2023. <https://www.forbes.com/councils/forbesbusinesscouncil/2023/06/02/the-coming-ai-revolution-in-home-care/>. Accessed Jul 3, 2025.
11. Tyagi A. How AI can improve outcomes in clinical home care. *Forbes Technology Council*. April 3, 2024. <https://www.forbes.com/councils/forbestechcouncil/2024/04/03/how-ai-can-improve-outcomes-in-clinical-home-care/>. Accessed Jul 3, 2025.
12. Nguyen TD, Whaley CM, Simon K, et al. Adoption of artificial intelligence in the health care sector. *JAMA Health Forum*. 2025;6(11):e255029. <https://doi.org/10.1001/jamahealthforum.2025.5029>.
13. Baumann S, Stone RT, Abdelall E. AI-enabled sensing and monitoring systems for healthcare applications. *Sensors*. 2025;25(19):6067. <https://doi.org/10.3390/s25196067>. Accessed May 4, 2026. <https://www.mdpi.com/1424-8220/25/19/6067>.
14. Solano-Kamaiko IR, Tan M, Yang I, et al. "This is eye opening," raising awareness of home care workers' health and wellbeing via activity tracking. In: *Proceedings of the CSCW Conference on Computer-Supported Cooperative Work and Social Computing (CSCW '25); October 18-22, 2025; Bergen, Norway*. <https://iansolano.com/assets/documents/Activity-Tracking-For-HCWs-CSCW25.pdf>. Accessed Jul 3, 2025.
15. CareConnect. CareConnect: caregiver recruitment, engagement, and retention. <https://careconnectmobile.com/>. Accessed Jul 3, 2025.
16. Careswitch Inc. AI assistant | Careswitch. <https://www.careswitch.com/ai-assistant>. Accessed Jul 3, 2025.
17. Sensi.AI. Sensi: 24/7 AI care intelligence copilot for caregivers. <https://www.sensi.ai/>. Accessed Jul 3, 2025.
18. Cingolani M, Scendoni R, Fedeli P, Cembrani F. Artificial intelligence and digital medicine for integrated home care services in Italy: Opportunities and limits. *Front Public Health*. 2023;10:1095001. Published 2023 Jan 5. <https://doi.org/10.3389/fpubh.2022.1095001>.
19. Glasby J, Litchfield I, Parkinson S, et al. New and emerging technology for adult social care - the example of home sensors with artificial intelligence (AI) technology. *Health Soc Care Deliv Res*. 2023;11(9):1-64. <https://doi.org/10.3310/HRW4281>.
20. Karnehed S, Larsson I, Petersson L, Erlandsson LK, Tyskbo D. Navigating artificial intelligence in home healthcare: challenges and opportunities in nursing wound care. *BMC Nurs*. 2025;24(1):660. Published 2025 Jun 19. <https://doi.org/10.1186/s12912-025-03348-7>.
21. Kuo EF, Cho J, Olaye I, Delgado D, Dell N, Sterling MR. Understanding the technological landscape of home health aides: scoping literature review and a landscape analysis of existing mHealth apps. *J Med Internet Res*. 2022;24(11):e39997. Published 2022 Nov 11. <https://doi.org/10.2196/39997>.
22. Dröge K. noScribe: cutting-edge AI technology for automated audio transcription (Version 0.6.2) [computer software]. GitHub. <https://github.com/kaixxx/noScribe>. Accessed Jul 3, 2025.
23. ATLAS.ti Scientific Software Development GmbH. *ATLAS.ti: the #1 software for qualitative data analysis* [computer software]. <https://atlasti.com/>. Accessed Jul 3, 2025.
24. Kuckartz U. *Qualitative Text Analysis: A Guide to Methods, Practice & Using Software*. SAGE Research Methods. <https://methods.sagepub.com/book/mono/qualitative-text-analysis/toc>. Accessed Jul 16, 2025.
25. Delve, Ho L, Limpaecher A. How to do structural coding. Delve Blog. Published October 2, 2024. <https://delvetool.com/blog/structural-coding>. Accessed Jul 16, 2025.
26. Braun V, Clarke V. Thematic analysis. *APA Handbook of research methods in psychology*. Vol. 2: research designs: quantitative, qualitative, neuropsychological, and biological. *Am Psychol Assoc*. 2012:57-71.
27. Neves BB, Omori M, Petersen A, Vered M, Carter A. Navigating artificial intelligence in care homes: Competing stakeholder views of trust and logics of care. *Soc Sci Med*. 2024;358:117187. <https://doi.org/10.1016/j.socscimed.2024.117187>.
28. Wang B, Asan O, Zhang Y. Shaping the future of chronic disease management: Insights into patient needs for AI-based homecare

- systems. *Int J Med Inform.* 2024;181:105301. <https://doi.org/10.1016/j.ijmedinf.2023.105301>.
29. **Bensson-Ravunniarath M, Bryan Ringel J, Avgar A, et al.** Having a say matters: the association between home health aides' voice and job satisfaction. *Risk Manag Healthc Policy.* 2023;16:1791–1800. Published 2023 Sep 8. <https://doi.org/10.2147/RMHP.S420207>.
30. **Lai D, Cloyes KG, Clayton MF, et al.** WE'RE THE EYES AND THE EARS, BUT WE DON'T HAVE A VOICE: PERSPECTIVES OF HOSPICE AIDES. *J Hosp Palliat Nurs.* 2018;20(1):47–54. <https://doi.org/10.1097/NJH.0000000000000407>.
31. **Campbell B, Thomson H, Slater J, Coward C, Wyatt K, Sweeney K.** Extracting information from hospital records: what patients think about consent. *Qual Saf Health Care.* 2007;16(6):404–408. <https://doi.org/10.1136/qshc.2006.020313>.
32. Health Insurance Portability and Accountability Act of 1996. Pub L No. 104–191, 110 Stat 1936.
33. Centers for Medicare & Medicaid Services. Electronic Visit Verification (EVV). Medicaid.gov. <https://www.medicaid.gov/medicaid/home-community-based-services/guidance/electronic-visit-verification-evv/index.html>. Accessed Jul 16, 2025.
34. World Health Organization. WHO outlines considerations for regulation of artificial intelligence for health. *WHO.int.* Published October 19, 2023. <https://www.who.int/news/item/19-10-2023-who-outlines-considerations-for-regulation-of-artificial-intelligence-for-health>. Accessed Jul 3, 2025.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.